On the Scattering-Angle Dependence of the Spectral Consistency of Ice Cloud Optical Thickness Retrievals Based on Geostationary Satellite Observations

Dongchen Li, Masanori Saito, Ping Yang, *Fellow, IEEE*, Norman G. Loeb, William L. Smith Jr., and Patrick Minnis

***Abstract*— Visible-near infrared (VIS-NIR) and thermal infrared (TIR) methods have long been used for ice cloud property retrievals based on satellite observations. Both retrieval methods are sensitive to the assumed ice particle models, which can significantly impact the accuracy of the retrieved microphysical and radiative properties of ice clouds. The two-habit model (THM) is considered a suitable ice particle model for passive remote sensing of global ice clouds, as confirmed by spectral consistency in cloud optical thickness (COT), which refers to the agreement between VIS-NIR and TIR COT retrievals. However, the ratio of COTs retrieved from these two methods using the THM varies with the scattering angle, indicating the potential influence of atmospheric and cloud properties on the spectral consistency of the inferred COT. The present study investigates potential factors affecting the angular dependence of the COT ratio based on observations made by the Advanced Baseline Imager (ABI) sensors on the 17th Geostationary Operational Environmental Satellite (GOES-17). For optically thin clouds (COT < 1) illuminated at a solar zenith angle (SZA) < 20°, the heterogeneous particle combination of mixed-phase clouds is the dominant factor that produces an angular-dependent negative bias in COT spectral consistency. At greater SZAs, cloud 3-D radiative effects are presumed to be a dominant factor. Negative biases in the cloud top temperature (CTT) and sea surface temperature contribute to negative biases in the COT ratio. This study suggests that the angular dependence of the COT ratio could help identify mixed-phase clouds and estimate their impact on the spectral consistency in retrieving ice cloud COT.**

***Index Terms*— Cloud optical thickness (COT), ice cloud, mixed-phase cloud, spectral consistency, thermal infrared (TIR) retrieval method, visible-near infrared (VIS-NIR) retrieval method.**

I. Introduction

ICE clouds play a crucial role in Earth's climate system through their regulation of the atmospheric radiative energy budget [1]. However, a fundamental understanding of the microphysical and radiative properties of ice clouds is still limited. Cloud optical thickness (COT) and cloud effective radius (CER) are two of the most important cloud properties because of their substantial impact on solar reflectance and cloud microphysical processes. The uncertainties associated with these cloud parameters in numerical models lead to inaccuracy in weather and climate predictions, while they also act as contributing factors to uncertainty in climate change projections [2]. Therefore, it is essential to continuously monitor ice clouds on a global scale and to examine the microphysical and optical properties that are useful for validating ice cloud schemes in climate models.

Space-based observations provide valuable opportunities to investigate ice cloud properties with high temporal and spatial resolutions. Passive and active remote sensing measurements have been employed in numerous studies to infer important cloud properties, such as COT, CER, cloud top temperature (CTT), and cloud thermodynamic phase [3], [4], [5], [6], [7], [8], [9], [10], [11], [12]. This study utilizes the Advanced Baseline Imager (ABI) on board the Geostationary Operational Environmental Satellites-17 (GOES-17), operated by NOAA and NASA. The ABI sensor is a state-of-the-art 16-band radiometer with channels covering the visible (VIS), near-infrared (NIR), and thermal infrared (TIR) portions of the electromagnetic spectrum. It provides high temporal resolutions ranging from 30 s to 15 min and spatial resolutions from 0.5 to 2 km.

Two retrieval approaches are commonly used to infer COT and CER based on typical spaceborne passive narrowband measurements. One is the VIS-NIR retrieval method, referred to as the Nakajima–King method [13], which is based on a pair of reflectances in shortwave channels. The other is the TIR retrieval method, known as the split-window method [14], which relies on the brightness temperature observed in the infrared window region [7], [15]. Since ice clouds have different optical properties with different ice crystal shapes [16], [17], the assumed ice crystal shapes have significant impacts on the retrieved COT using either VIS-NIR or TIR retrieval methods [18], [19]. As a result, it is important to have an ice particle shape model that can accurately represent natural ice clouds.

Manuscript received 11 August 2023; revised 13 October 2023; accepted 31 October 2023. Date of publication 9 November 2023; date of current version 27 November 2023. This work was supported in part by Texas A&M University internal funds under Grant 02-132532-00095 and in part by NASA under Grant 80NSSC22K1890. *(Corresponding author: Dongchen Li.)*

Dongchen Li, Masanori Saito, and Ping Yang are with the Department of Atmospheric Sciences, Texas A&M University, College Station, TX 77843 USA (e-mail: lidc@tamu.edu).

Norman G. Loeb and William L. Smith Jr. are with the Science Directorate, NASA Langley Research Center, Hampton, VA 23681 USA.

Patrick Minnis is with the Science Directorate, NASA Langley Research Center, Hampton, VA 23681 USA, and also with Analytical Mechanics Associates Inc., Hampton, VA 23681 USA.

Digital Object Identifier 10.1109/TGRS.2023.3331970

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:58 UTC from IEEE Xplore. Restrictions apply.

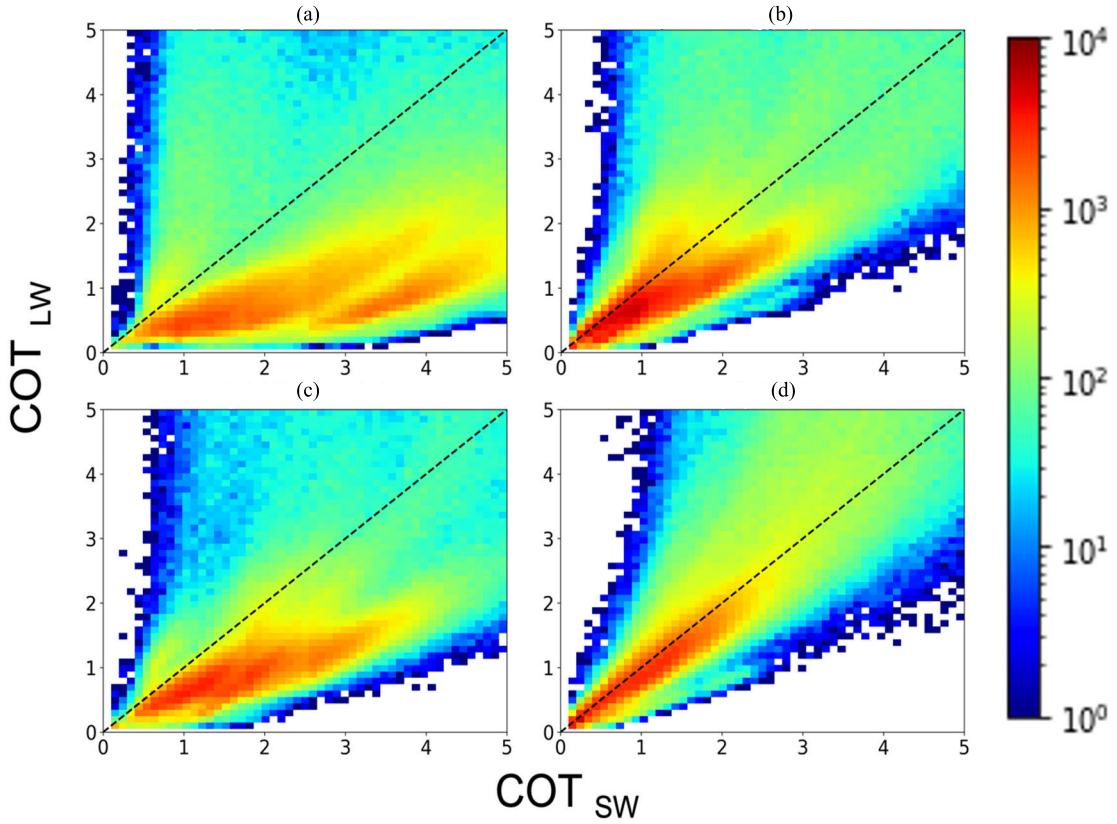


Fig. 1. Comparison of retrieved COT from the VIS-NIR and TIR methods based on (a) spherical ice model, (b) five-plates aggregate model, (c) HBR model, and (d) THM. The COT_{SW} and COT_{LW} axes represent COT retrievals performed with the VIS-NIR and TIR methods, respectively.

Based on in situ measurements and simulation studies of ice particle shapes, it is well-known that ice clouds are composed of nonspherical ice crystals [20]. Several ice particle optical property models have been developed and improved to accurately represent the microphysical and optical properties of ice clouds [21], [22]. This study uses an improved two-habit model (THM) built upon its previous version reported by Loeb et al. [19]. The improved THM comprises a mixture of a 60-particle ensemble of irregularly shaped single hexagonal columns and an aggregate of 20 irregularly shaped hexagonal columns with the habit distribution shown in Coy et al. [23].

Here, spectral consistency means that retrievals of COT based on VIS-NIR and TIR observations of the same cloud agree well [24], [25], [26]. It should be pointed out that spectral consistency may not be achieved in retrieving cloud-effective particle size because different spectral channels are sensitive to different portions of a cloudy layer that may be vertically inhomogeneous. The spectral consistency serves to validate ice optical property models used in remote sensing techniques. Fig. 1 compares COT values from the VIS-NIR and TIR retrievals based on different ice particle shape assumptions using GOES-17 observations from 20:00 to 20:50 UTC on September 23, 2019. It is evident that spectral consistency is achieved for THM [see Fig. 1(d)], while the other ice particle models, such as spherical ice [see Fig. 1(a)], five-plate aggregates [see Fig. 1(b)], and hollow bullet rosettes (HBRs) in Fig. 1(c), do not exhibit the same level of spectral consistency. Therefore, THM is considered a suitable ice particle model for passive remote sensing of global ice clouds. As the COT increases, TIR signals become less sensitive to the lower portion of the cloud. Consequently, the retrieval uncertainties from the TIR method increase and the TIR COT retrievals are systematically lower than their VIS-NIR counterparts.

However, several other factors can introduce biases and uncertainties into the retrievals based on the two methods, consequently impacting the spectral consistency of COT. Since the remote sensing techniques assume that clouds are plane-parallel and homogeneous in every satellite pixel, the reflectance and brightness temperature of a cloudy pixel could be influenced by subpixel-scale cloud inhomogeneity, which could lead to uncertainties in the cloud optical property retrievals from both VIS-NIR and TIR methods [27], [28], [29], [30], [31]. Approximately 20% of the identified cloudy pixels correspond to multilayer cloud systems, leading to a significant impact on the retrieval of cloud optical properties [32]. This impact is particularly pronounced for multilayer clouds where the COT of the upper ice cloud layer is <5 [33]. The three-dimensional (3-D) structure of clouds can lead to shadowing and varying illumination effects, resulting in errors in the retrievals of COT ranging from -5.6% to 19.6% and CER ranging from -2.9% to 8.0% [34]. When COT varies horizontally from pixel to pixel, the measured reflectance for a thin cirrus pixel is often a combination of the reflectance of the observed cloud and a surface reflectance corresponding to another cloud having a COT different from that of the observed cloud. This can alter the observed reflectance by up to 50% or more depending

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:58 UTC from IEEE Xplore. Restrictions apply.

on the viewing and illumination angles as well as the COTs of the observed and shading clouds [35]. In addition, several previous studies also demonstrate that thin cirrus clouds and mixed-phase clouds are challenging to classify correctly due to their complex optical properties and the limitations of current cloud top phase retrieval algorithms [36], [37]. As a result, retrievals of cloud optical properties could be biased due to the misidentification of the cloud phase [38]. Although spectral consistency of COT is achieved for THM [see Fig. 1(d)], there is still a significant level of uncertainty regarding other potential physical processes that could influence the degree of spectral consistency.

In this study, we examine factors that potentially affect the uncertainty in cloud property retrievals from both VIS-NIR and TIR retrieval methods, including possible contamination by mixed-phase clouds, CTT biases, and sea surface temperature (SST) biases. In addition, we discuss the potential influences of cloud 3-D radiative and cloud heterogeneity effects on COT retrievals obtained from the VIS-NIR method. The remainder of this article is as follows: Section II describes the data and selection criteria, the mixed-phase bulk property calculations, the VIS-NIR and TIR retrieval systems, and the synthetic simulation procedure. Section III presents the results obtained from GOES-17 observations. Section IV comprehensively analyzes and interprets the results. Section V summarizes the major conclusions derived from this study.

TABLE I
PIXEL DATA ACCEPTANCE CRITERIA

Variable name	Threshold for acceptance
Cloud top phase	Ice
SZA	$< 60°$
VZA	$< 80°$
Surface type mask	Ocean
Day/Night flag	Day
Sunglint angle	$> 30°$
k (reflectance ratio)	< 2.5

II. DATA AND METHODOLOGY

A. Data and Pixel Selection Criteria

This study uses two days of the GOES-17 ABI full-disk operational products on March 20, 2019 and September 23, 2019, the spring and fall equinoxes. Since GOES-17 orbits at a fixed altitude (about 35 880 km) over the equator, more pixels with lower solar zenith angles (SZAs) are available in both hemispheres than on other days of the year. GOES-17 ABI full-disk operational products cover a range from 81.3282° South to 81.3282° North in latitude, and from 141.9005° East to 55.9005° West in longitude. Reflectances at 0.64 μm (band 2) and 2.24 μm (band 6) with 0.5- and 2-km resolution from GOES-17 are used to perform VIS-NIR retrievals. Since the resolution of band 2 is finer than band 6, the band 2 reflectance data are pixel-averaged to achieve a 2-km resolution [28].

This study uses GOES-17 ABI full-disk brightness temperatures with 2-km resolution at 8.4 μm (band 11), 10.3 μm (band 13), 11.2 μm (band 14), and 12.3 μm (band 15) for the TIR retrievals. Ancillary data used in the TIR retrieval include the GOES-17 CTT product, used as the *a priori* value to retrieve CTT. The Modern-Era Retrospective Analysis for Research and Applications, version 2 (MERRA-2), produced by NASA Global Modeling and Assimilation Office (GMAO), provides comprehensive global reanalysis products with $0.5° \times 0.625°$ horizontal resolution. M2I3NPASM is an instantaneous 3-D 3-h data collection in MERRA-2, which contains assimilations of meteorological parameters at 42 pressure levels, including vertical profiles of geopotential height, pressure, temperature profile, and water vapor content. Total precipitable water (w) can be obtained from M2I1NXASM, which is a 2-D hourly data collection in MERRA-2. Level 4 SST data with 0.25° resolution produced by the Jet Propulsion Laboratory (JPL) Physical Oceanography Distributed Active Archive Center (PO-DAAC) are employed in this TIR retrieval system. The diurnal variation of SST is not considered in this study. These data were linearly interpolated horizontally to match the spatial resolution of the GOES-17 image and subsequently utilized as ancillary data for the TIR retrieval system.

To enhance the quality and accuracy of the retrieval results, we exclusively consider pixels that satisfy the criteria summarized in Table I. We utilize the GOES-17 cloud top phase product to identify pixels classified as ice cloud coverage. Specifically, we restrict our analysis to pixels over oceanic regions due to the relatively low surface albedo compared to land, and invariant surface emission, which is set to 0.99 in the TIR channels. The sun-view geometries are used to calculate the sunglint angle [28]. To mitigate retrieval failures, we impose a constraint on the SZA and viewing zenith angle (VZA), limiting them to values below 60° and 80°, respectively. We also filter out pixels having sunglint angles less than 30° to reduce the sunglint effect [39].

In addition, GOES-17 reflectances in the 0.64- and 1.37-μm (band 4) are used to mitigate the effect of multilayer clouds. The band-4 reflectance (R_{band4}) is sensitive to ice clouds, while the band-2 reflectance (R_{band2}) includes the contribution from ice clouds, lower layer clouds, and surface reflectance (R_{sfc}),

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:58 UTC from IEEE Xplore. Restrictions apply.

so we can detect pixels covered by multilayer clouds if we exclude surface contamination in the band-2 reflectance. The ocean surface albedo (α_{alb}) can be calculated from SZA [40], and converted into reflectance using the following equation:

$$R_{\text{sfc}} = \alpha_{\text{alb}} \exp\left(-\frac{\tau_{\text{sw}}}{\mu} - \frac{\tau_{\text{sw}}}{\mu_0}\right) \tag{1}$$

where τ_{sw} is the COT retrieved by the VIS/NIR method, and μ and μ_0 are the cosines of VZA and SZA, respectively. The reflectance ratio (k) is defined as follows:

$$k = \frac{R_{\text{band2}} - R_{\text{sfc}}}{R_{\text{band4}}}. \tag{2}$$

If there is no multilayer cloud effect, k should be close to one due to a smaller spectral dependence of reflectance from the ocean surface. Therefore, we filter out pixels with $k \geq 2.5$ to mitigate the multilayer cloud effect.

B. Mixed-Phase Bulk Optical Property Calculation

To retrieve cloud optical properties based on GOES-17 observations, it is crucial to calculate the absorption and scattering properties of clouds as bulk optical properties using the specific GOES-17 spectral channels. Liquid clouds consist of spherical liquid droplets, allowing for the computation of their single-scattering properties with the Lorenz–Mie theory [41]. Because ice clouds consist of a variety of nonspherical crystals, the single-scattering properties of ice crystals are obtained from the THM in this study. Using single-scattering properties and assuming a Gamma distribution for both ice and liquid particle sizes, we can calculate the bulk optical properties of ice and liquid clouds, including bulk extinction efficiency $\langle Q_{\text{ext}} \rangle$, bulk scattering efficiency $\langle Q_{\text{sca}} \rangle$, bulk single scattering albedo $\langle \omega \rangle$, bulk asymmetry factor $\langle g \rangle$, and bulk phase function $\langle P_{11} \rangle$ [42]. The effective radius (r_{eff}) of ice and liquid particles is defined as follows:

$$r_{\text{eff}} = \frac{3}{4}\frac{\langle V \rangle}{\langle A \rangle} \tag{3}$$

where $\langle V \rangle$ is the bulk volume of particles and $\langle A \rangle$ is the bulk projected area of particles.

A mixed-phase cloud is treated as a mixture of the volumes of ice and liquid clouds within the vertical column of a pixel. To calculate the bulk properties of a mixed-phase cloud, the liquid volume fraction (f_V^l) is defined as follows:

$$f_V^l = \frac{f_N^l \langle V^l \rangle}{f_N^l \langle V^l \rangle + f_N^i \langle V^i \rangle} \tag{4}$$

where f_N represents of the particle number fraction. In this section, superscripts "l," "i," and "m" represent liquid, ice, and mixed-phase, respectively.

Since $\langle Q_{\text{ext}} \rangle$ and $\langle g \rangle$ are dependent on the projected area and extinction cross section of particles, and $\langle g \rangle$ and $\langle P_{11} \rangle$ are mainly influenced by the scattering cross sections of particles, the liquid volume fraction needs to be weighted by particle bulk projected area, extinction cross section, and scattering cross section using the following equations:

$$f_A^l = \frac{\frac{f_V^l}{r_{\text{eff}}^l}}{\frac{f_V^l}{r_{\text{eff}}^l} + \frac{1-f_V^l}{r_{\text{eff}}^i}} \tag{5}$$

$$f_{\text{ext}}^l = \frac{\frac{\langle Q_{\text{ext}l} \rangle f_V^l}{r_{\text{eff}}^l}}{\frac{\langle Q_{\text{ext}}^l \rangle f_V^l}{r_{\text{eff}}^l} + \frac{\langle Q_{\text{ext}}^i \rangle (1-f_V^l)}{r_{\text{eff}}^i}} \tag{6}$$

$$f_{\text{sca}}^l = \frac{\frac{\langle Q_{\text{sca}}^l \rangle f_V^l}{r_{\text{eff}}^l}}{\frac{\langle Q_{\text{sca}}^l \rangle f_V^l}{r_{\text{eff}}^l} + \frac{\langle Q_{\text{sca}}^i \rangle (1-f_V^l)}{r_{\text{eff}}^i}} \tag{7}$$

where f_A^l, f_{ext}^l, and f_{sca}^l are liquid volume fractions weighted by bulk projected area, extinction cross section, and scattering cross section. The above formulas are consistent with Yang et al. [43]. Based on the weighted volume fraction of liquid droplets, we can calculate the bulk optical properties of mixed-phase clouds as follows:

$$\langle Q_{\text{ext}}^m \rangle = \langle Q_{\text{ext}}^l \rangle f_A^l + \langle Q_{\text{ext}}^i \rangle (1 - f_A^l) \tag{8}$$

$$\langle \omega^m \rangle = \langle \omega^l \rangle f_{\text{ext}}^l + \langle \omega^i \rangle (1 - f_{\text{ext}}^l) \tag{9}$$

$$\langle g^m \rangle = \langle g^l \rangle f_{\text{sca}}^l + \langle g^i \rangle (1 - f_{\text{sca}}^l) \tag{10}$$

$$\langle P_{11}^m \rangle = \langle P_{11}^l \rangle f_{\text{sca}}^l + \langle P_{11}^i \rangle (1 - f_{\text{sca}}^l). \tag{11}$$

C. Retrieval Methods

In the forward model of the VIS-NIR retrieval system, an adding-doubling radiative transfer model is used to construct a lookup table (LUT) [44], which contains various combinations of COT, CER, and the corresponding reflectance. By utilizing the pre-calculated LUT, the retrieval process becomes computationally efficient as it does not require iterative radiative transfer computations. The TIR retrieval method relies on the differences in TIR radiation emitted by the clouds to estimate CTT, COT, and CER. It is more sensitive to optically thin clouds compared to the VIS-NIR retrieval method and can be used for both daytime and nighttime retrievals. In this study, we use a single-layer radiative transfer model based on the two-stream approximation in the forward model of the TIR retrieval system [45], [46], to calculate the brightness temperature of each GOES-17 infrared band.

Using the forward model in the VIS-NIR and TIR retrieval systems, the Nakajima–King and split-window diagrams are computed for ice, mixed-phase, and liquid cloud cases. Fig. 2 illustrates how the mixed-phase cloud optical properties affect cloud radiative signals. The Nakajima–King diagrams are generated with SZA = 30°, VZA = 60°, and relative azimuth angle (RAA) of 20°. The split-window diagrams are computed using VZA = 60°, CTT = 250 K, SST = 297 K, precipitable water = 17.82 mm, and a constant surface emissivity (E_{sfc}) of 0.99 in each infrared band. The bulk optical properties of a mixed-phase cloud are calculated based on (8)–(11) with a liquid volume fraction of 0.5. Comparisons among Fig. 2(a)–(c) reveal that the optical properties of mixed-phase clouds are significantly influenced by those of liquid clouds in VIR and NIR channels. Fig. 2(b) shows a severe nonorthogonal correlation between COT and CER when COT < 8. This indicates the challenge of obtaining the optimal solutions using the VIS-NIR retrieval method based on the mixed-phase model. Although the optical properties of liquid droplets still have a nonnegligible impact on the CER of mixed-phase clouds at the TIR wavelengths, the COT of mixed-phase clouds is less affected by liquid droplets compared to the VIS and NIR

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:58 UTC from IEEE Xplore. Restrictions apply.

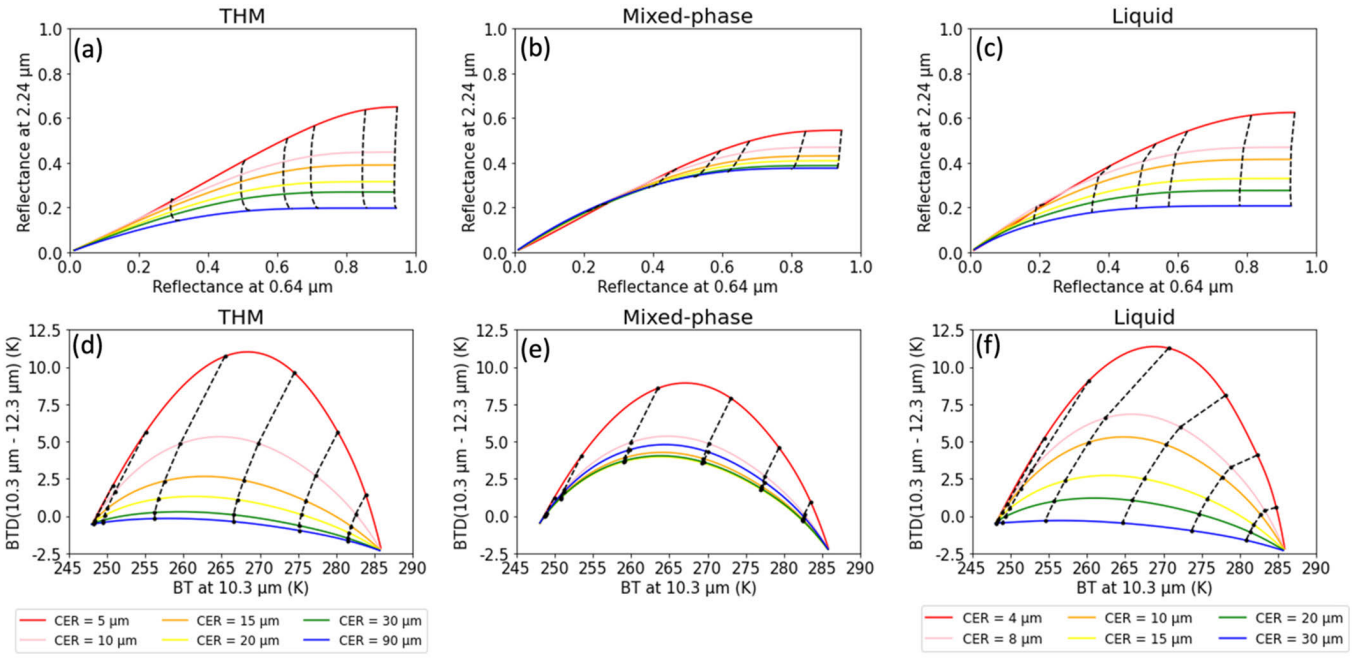


Fig. 2. (a)–(c) Nakajima–King and (d)–(f) split-window diagrams for THM, mixed-phase, and liquid model. In the Nakajima–King diagrams, dashed lines show constant COT at 5, 8, 12, 16, 32, and 64 from left to right. In the split-window diagrams, dashed lines show constant COT at 6, 4, 2, 1, 0.5, and 0.2 from left to right. The CER for THM is the same in both Nakajima–King and split-window diagrams, as is the CER for the liquid model. (b) and (e) CER of mixed-phase cases (CERmix) is a combination of liquid droplets with a CER of 10 μm and ice particles with a CER consistent with the ice cloud case. BT and BTD represent the brightness temperature and brightness temperature difference, respectively.

channels [shown in Fig. 2(d)–(f)]. This is due to the relatively lower absorption and scattering of liquid droplets in the TIR spectrum compared to ice particles. It implies less interaction with the incoming TIR radiation, resulting in lower sensitivity of liquid droplets at TIR channels in a mixed-phase cloud. As a result, the optical properties of liquid droplets have a stronger influence on the mixed-phase cloud optical properties in the VIS-NIR retrieval method, while their impact diminishes in the TIR technique.

The optimal estimation algorithm is applied in the inversion part of both VIS-NIR and TIR retrieval systems [47], which can obtain an optimal solution by minimizing the cost function (see [9]). The cost function includes the measurement results either from the GOES-17 observations or calculated from the cloud model parameters using the forward models. It also includes the state vectors of cloud parameters along with their corresponding *a priori* values, as well as the error covariance matrix for both measurements and *a priori* values. The VIS-NIR retrieval method uses *a priori* values of 5.62 for COT and 19.95 μm for CER in the state vectors, with the retrieval parameter space ranging from 0.1 to 150 for COT and from 1 to 100 μm for CER. In the TIR retrieval method, the GOES-17 CTT operational product is utilized as the *a priori* value for CTT with a range of 180–230 K. In addition, the state vectors of COT and CER have *a priori* values of 3.0 and 30 μm, respectively, with ranges of 0.3–8 for COT and 5–100 μm for CER.

III. Observational Results

Fig. 3 shows the COT ratios across the scattering angle range obtained from the VIS-NIR retrievals and TIR retrievals for nine different combinations of CTT and COT. The scattering angle is defined as the angle between the incident and scattered radiation beams. The COT ratio is defined by the COT from the Nakajima–King method divided by that from the split-window method. If the assumed ice particle model has optical properties that are consistent with those of the particles in natural clouds, and the signals received by the satellite are solely from clouds, we should obtain a flat line without the scattering-angle dependence. However, as seen in Fig. 3(a), the median COT ratio displays a significant angular dependence when the clouds are optically thin (COT $<$ 1) with very cold CTT (200 K $<$ CTT $<$ 210 K). This indicates that the retrieval of COT could still be biased, even though the best ice property model currently available is applied in the retrieval process.

As COT becomes larger, the COT retrieval uncertainty diminishes in both the VIS-NIR and TIR methods. This uncertainty reduction is particularly pronounced for the TIR method, which has a strong sensitivity to optically thin clouds with COT $>$ 1. Therefore, as the COT increases, the uncertainty of the COT ratio substantially decreases. The accuracy of the TIR COT retrieval diminishes with increased CTTs due to the reduced thermal contrast between the cloud and the surface, thereby contributing to high-cloud CTT uncertainties. In addition, high and optically thin ice clouds are typically found in the upper level of multilayer cloud systems. As the CTT increases, the occurrence of multilayer cloud systems might decrease, leading to reduced retrieval biases and uncertainties in the VIS-NIR method. This decrease in multilayer cloud occurrence might contribute to the suppression of the variation in the COT ratio with the scattering angle.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:58 UTC from IEEE Xplore. Restrictions apply.

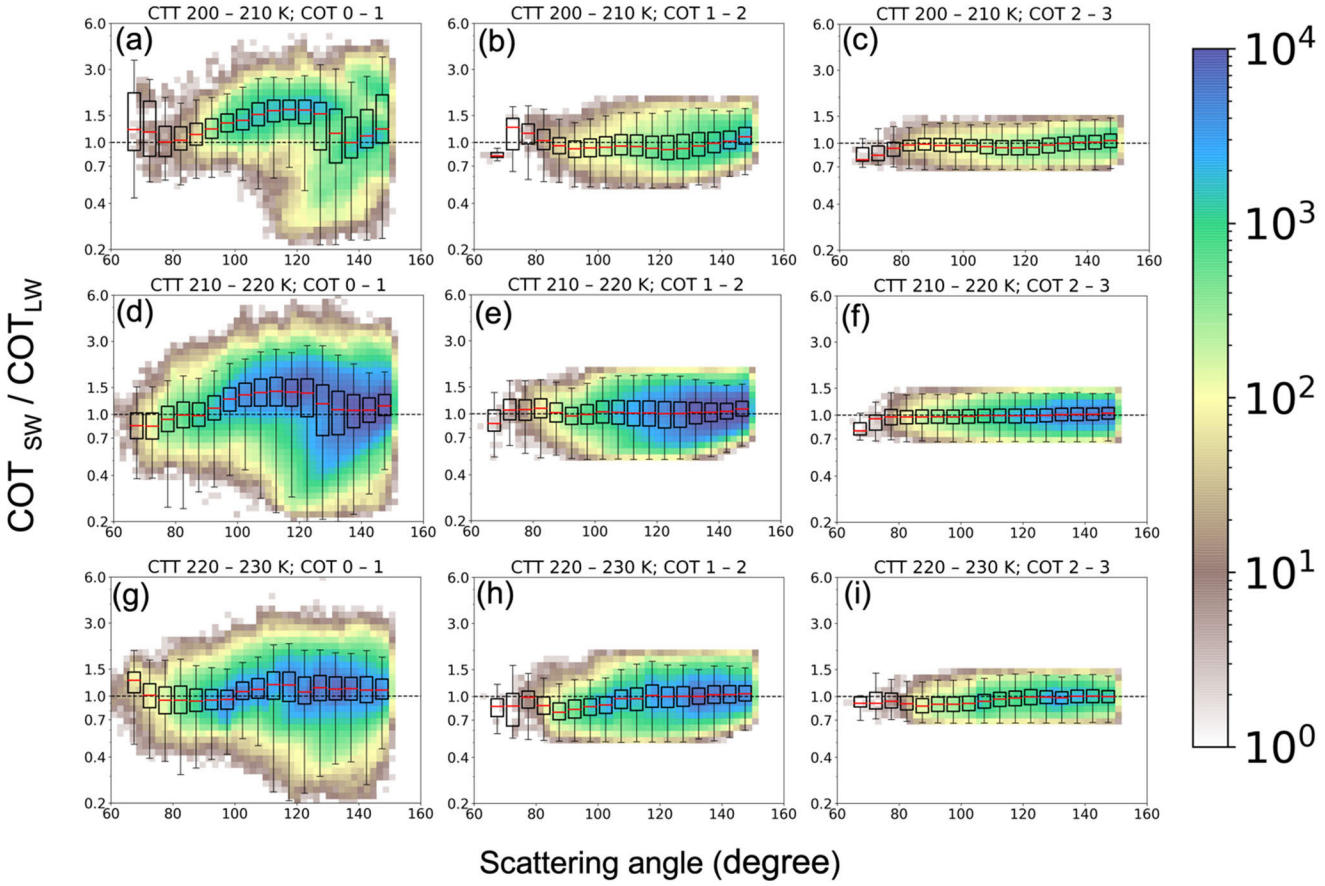


Fig. 3. (a)–(i) Two-dimensional histogram and box plots illustrating the distribution of the COT ratio as a function of scattering angle, categorized into three CTT ranges (200–210 K, 210–220 K, 220–230 K) and three COT ranges (0–1, 1–2, 2–3). The red horizontal lines represent the median values of the COT ratio.

To narrow down the potential factors contributing to the angular dependence of the COT ratios, we further divide the cases shown in Fig. 3(a)–(c) into two SZA ranges for each: 0°–20° and 20°–60°. Fig. 4(a) shows an obvious increase in the COT ratio with a scattering angle from 120° to a peak around 140°. Most COT retrieval results from the VIS-NIR method are smaller than those from the TIR method, but in Fig. 4(d), when SZA $>$ 20°, the COT ratio is generally >1, indicating an overestimation of the COT retrievals in the VIS-NIR method compared to those in the TIR method. The uncertainties and variation of the COT ratio decrease as the COT increases within the same CTT range.

IV. Numerical Experiments and Discussion

Variation of the COT ratio with scattering angle may be influenced by several factors. The current cloud-top phase classification algorithm employed for GOES-17 relies mainly on empirical thresholds for CTT retrieved using multiple infrared wavelengths [48], which could introduce potential biases on the discrimination of cloud phase [49]. Thermal-based cloud top phase retrievals are sensitive to errors, particularly in detecting optically thin and broken clouds [50]. In addition, the optical property retrievals of thin ice clouds could be affected by liquid or mixed-phase clouds due to misidentification of the cloud phase and the inhomogeneity of clouds within the pixels. Although multilayer cloud filtering is applied in this study, this filtering method can only reduce the effect from the lower layer clouds to some extent. The retrieved optical properties of very thin and fragmented ice clouds with low CTT could still be subject to biases caused by the presence of underlying liquid or mixed-phase clouds. Therefore, the sharp increase of the COT ratio in the scattering angle range from 120° to 140° in Fig. 4(a) could imply contamination from liquid or mixed-phase clouds.

Any errors or uncertainties in $\langle P_{11} \rangle$ can introduce biases or inaccuracies in the retrieved COT. Fig. 5 demonstrates the impact of the presence of liquid droplets on $\langle P_{11} \rangle$ of ice particles. The $\langle P_{11} \rangle$ of THM is almost independent of the scattering angle between 60° and 160°. However, $\langle P_{11} \rangle$ of liquid droplets shows large variations in the near backscattering regions. As we calculate the ratio of $\langle P_{11} \rangle$ between THM and the mixed-phase model with various liquid volume fractions, the variation of the $\langle P_{11} \rangle$ ratio becomes larger as the liquid volume fraction increases. The $\langle P_{11} \rangle$ ratio of mixed-phase clouds increases in the scattering angle range from 110° to 140°, reaching its peak at a scattering angle near 140°, where a rainbow occurs. The angular dependence of the $\langle P_{11} \rangle$ ratio could contribute to the variation of the COT ratio with respect to the scattering angle.

To gain a deeper understanding of the mixed-phase cloud effect, a simulation experiment is designed using the data from Fig. 4(a)–(c). In this synthetic simulation, the mixed-phase

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:58 UTC from IEEE Xplore. Restrictions apply.

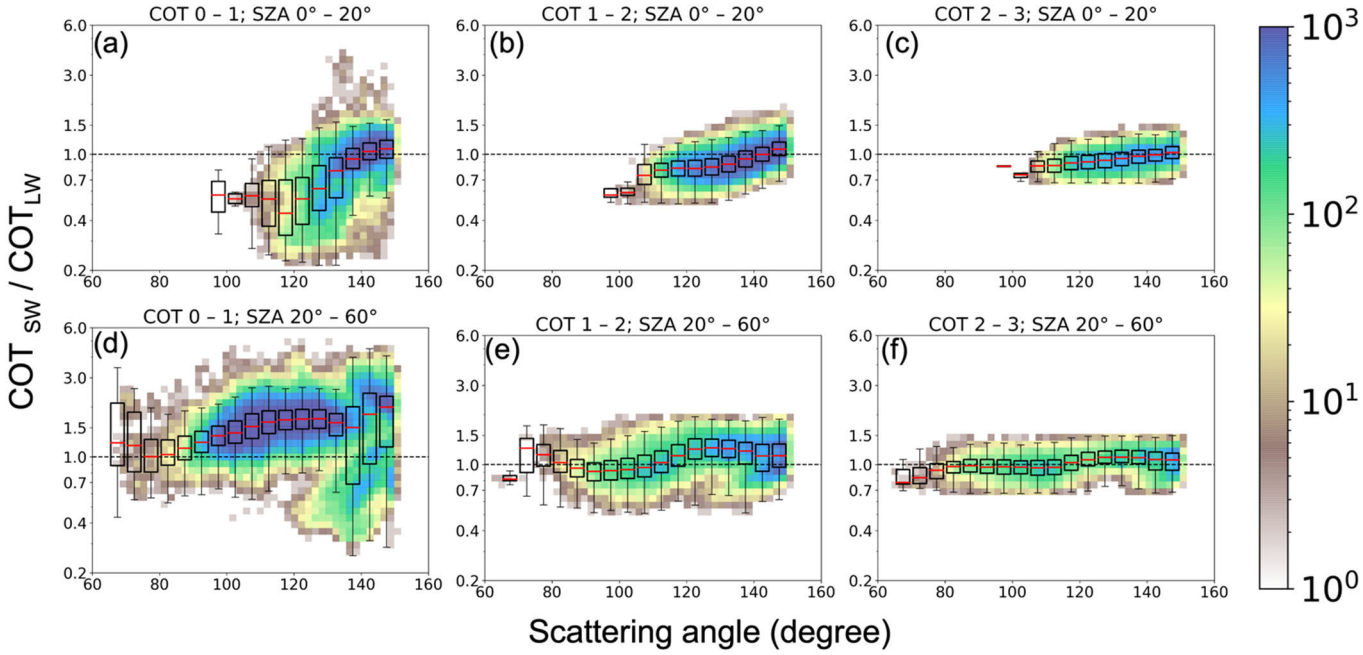


Fig. 4. (a)–(f) Similar to Fig. 3, but showing two-dimensional histogram and box plots illustrating the distribution of the COT ratio as a function of scattering angle within the same CTT range (200–210 K). The data are categorized into three COT ranges (0–1, 1–2, 2–3) and two SZA ranges (0–20°, 20–60°).

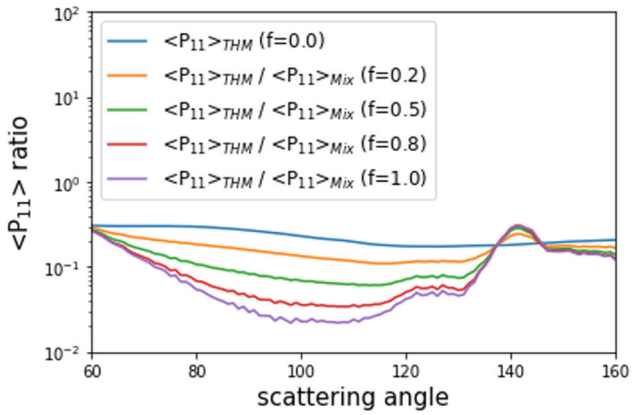


Fig. 5. $\langle P_{11}\rangle$ of THM and the $\langle P_{11}\rangle$ ratio of THM and the mixed-phase model with various liquid volume fractions in scattering angles 60°–160° at 0.64 μm wavelength. The CER of THM is 33.01 μm. The CER of the mixed-phase model (CER_{mix}) is obtained by combining the CER of 33.01 μm for ice particles and the CER of 10 μm for liquid droplets.

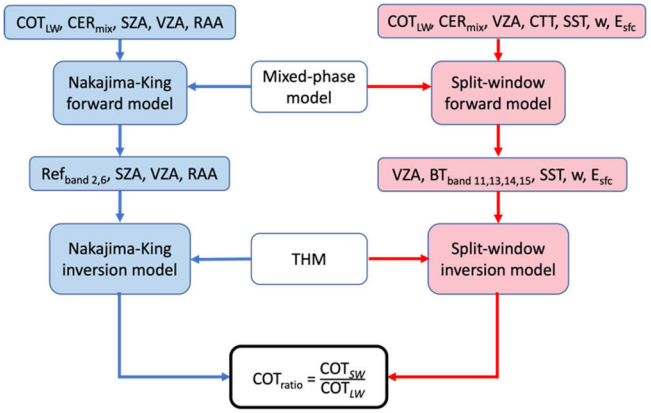


Fig. 6. Flowchart of the simulation process for evaluating the impact of a mixed-phase cloud on the spectral consistency of COT. Ref and BT represent the reflectance and brightness temperature, respectively.

model is obtained by combining the THM and liquid models with a volume fraction of 0.5. This means the mixed-phase model represents a cloud composed of an equal volume of ice particles and liquid droplets. The CER of liquid droplets is defined as 10 μm in the mixed-phase model. The flowchart depicted in Fig. 6 outlines the process of the numerical experiments for the mixed-phase cloud effect. Since the mixed-phase clouds have relatively less impact on the retrieval results from the TIR method as shown in Fig. 2(e), the COT retrieved from the TIR method is employed in the forward models in both the VIS-NIR and TIR retrieval systems. The solar-view geometries applied for this numerical experiment are calculated from the GOES-17 observations, following the same approach as the one applied in Fig. 8. In addition, the mixed-phase model is incorporated in the forward models to simulate the radiometric signals of solar reflectance and brightness temperature in GOES-17 ABI bands. Based on the simulated radiative signals obtained from the forward models, the simulated COT results can be retrieved using THM in the VIS-NIR and TIR methods, respectively.

Due to the inconsistent ice particle models assumed in the forward and inversion models, the COT retrieval results from the VIS-NIR and TIR methods include both optimal and nonoptimal solutions.

Using the simulated results of COT from the VIS-NIR and TIR retrieval methods, the variation of the simulated COT ratio can be obtained as the scattering angle changes, as shown in Fig. 7. Since the ice particles are more scattered than the mixed-phase particles, the retrieval results of COT are underestimated when using THM to retrieve mixed-phase clouds in the VIS-NIR retrieval method. As a result, most COT retrieval results from the TIR retrieval method are larger

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:58 UTC from IEEE Xplore. Restrictions apply.

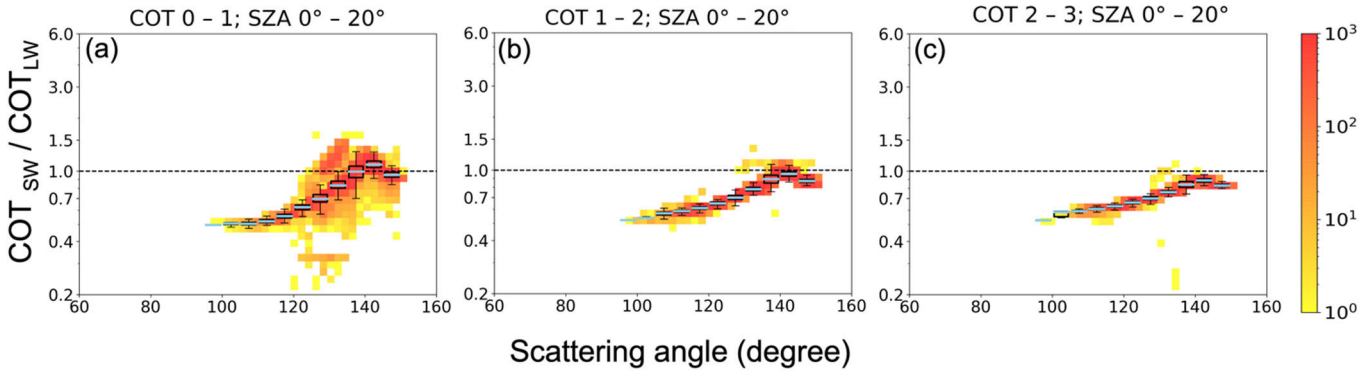


Fig. 7. Simulation results of COT ratio as a function of the scattering angle under the condition of SZA < 20° based on inconsistent ice particle models, with COT retrieval results from the TIR retrieval method ranging from (a) 0–1, (b) 1–2, and (c) 2–3. The blue horizontal lines represent the median values of the COT ratio.

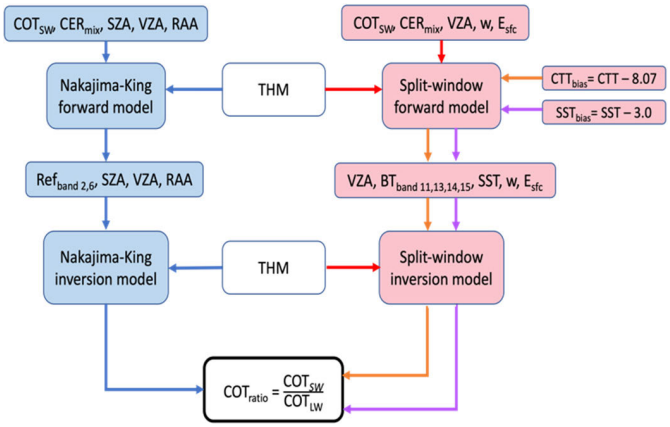


Fig. 8. Flowchart of the simulation process for evaluating the impact of CTT and SST on the spectral consistency of COT.

than those from the VIS-NIR retrieval method, as shown in both the observation [see Fig. 4(a)–(c) and simulation results (see Fig. 7). From Fig. 7(a), the variation of the median value of the simulated COT ratio displays a very similar feature as the observation results shown in Fig. 4(a), which is a strong indication of the impact from mixed-phase clouds. When the ice cloud is optically thin (COT < 1), the single scattering properties of ice particles play a significant role in the overall radiative signal. Therefore, the variation of the simulated COT ratio reflects changes in the ratio of $\langle P_{11} \rangle$ with respect to the scattering angle as shown in Fig. 5. Due to the complexity of typical tropical cloud systems, it is challenging to correctly identify the cloud thermodynamic phase and eliminate the multilayer cloud effect. However, evaluating the angular dependence of the COT ratio using both the VIS-NIR and TIR retrieval methods could provide a clue in diagnosing the presence of liquid or mixed-phase clouds.

The contribution of multiple scattering to the solar reflectance from clouds becomes significant as the COT increases, and therefore, the bulk asymmetry factor $\langle g \rangle$ has a relatively more pronounced impact on the radiative signal than $\langle P_{11} \rangle$ for optically thicker clouds. Since $\langle g \rangle$ is larger in a mixed-phase cloud than in an ice cloud, the COT retrieved from a mixed-phase cloud using the VIS-NIR method based on THM can be underestimated. Therefore, in Fig. 7(b) and (c), the maximum peak of the simulated COT ratio near the 140° scattering angle drops as COT increases. This decrease implies a transition from the $\langle P_{11} \rangle$ effect to the $\langle g \rangle$ effect. From Fig. 7(b) and (c), the simulated COT ratio becomes less variable with the scattering angle compared to Fig. 7(a). This can be attributed to larger contributions from multiple scattering that suppress the angular dependence associated with the single-scattering properties. In addition, the impact of $\langle g \rangle$ becomes dominant as COT increases, which is independent of the scattering angle. As a result, the simulated COT ratio shows relatively less dependence on the scattering angle as COT increases. This is consistent with the trend observed in the COT ratio retrieved based on the observation signals shown in Fig. 4(a)–(c). The COT ratios in Fig. 4(a) and (b) are slightly higher than the simulated COT ratio as COT is large. The difference could be due to the cloud 3-D radiative effect caused by a large VZA, which compensates for the negative biases in the COT ratio.

However, the mixed-phase cloud effect fails to explain the positive biases in the COT ratio shown in Fig. 4(d)–(f). Due to the complex vertical structure of clouds, potential nonmonotonic variations of temperature with height/pressure, and limited vertical resolution of atmospheric temperature profiles, large uncertainties, and biases could exist in the retrievals for broken clouds, thin cirrus clouds, and multilayer clouds [51], [52]. The SST retrieved from the TIR method could be regionally biased due to the restriction for clear-sky conditions [53]. Therefore, potential biases and uncertainties in CTT and SST could exist in the TIR retrieval process.

To understand the angularly dependent COT ratio in the case of SZA > 20°, potential biases in the TIR retrieval results are examined. We apply a consistent THM in both the forward and inversion models of the VIS-NIR and TIR retrieval systems, while biased CTT and SST values are separately introduced in the forward model of the TIR retrieval system for each simulation process (shown in Fig. 8). As CTT and SST have negligible impacts on the VIS-NIR retrieval method, COT retrieval results obtained from the VIS-NIR method are used in this simulation. By comparing the retrieval of CTT from geostationary satellite INSAT-3-D and Cloud-Aerosol Lidar and Infrared Pathfinder Satellite Observations (CALIPSOs), Lima et al. [54] report an overestimation in CTT retrieved by the TIR method for optically thin clouds. The CTT operational

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:58 UTC from IEEE Xplore. Restrictions apply.

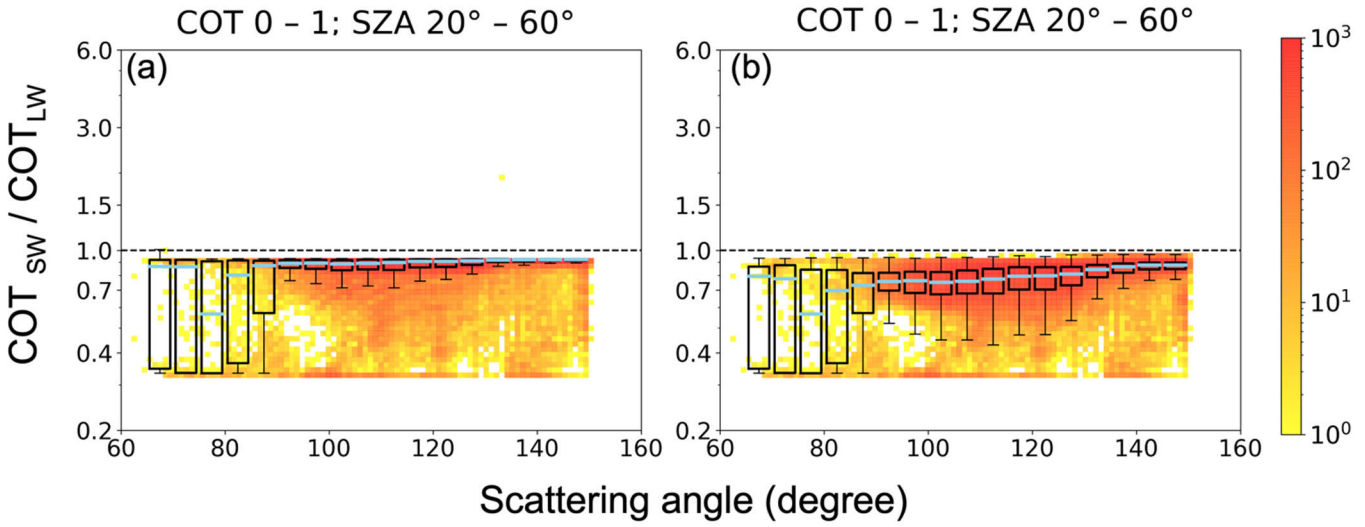


Fig. 9. Similar to Fig. 7, but showing the simulation results of the COT ratio as a function of scattering angle with SZA > 20° using a consistent ice particle model based on biased (a) CTT and (b) SST. This simulation result utilizes COT retrieved from the VIS-NIR retrieval method, which ranges from 0 to 1.

product from GOES-17 also exhibits a slight overestimation of CTT for ice clouds when compared with CALIPSO [55]. In addition, since SST retrieved from TIR is restricted to clear sky, midlatitude SST retrievals appear to be too warm [53]. To simulate the actual infrared signals received by GOES-17, we introduce negative biases of 8.07 and 3.0 K to CTT and SST, respectively, in the forward model of the TIR retrieval system.

Fig. 9 shows the COT ratios determined with the simulation process described in Fig. 8. The simulated COT ratios exhibit less variation across with scattering angle, resulting in small negative biases. Since the biases in CTT have less impact on the COT retrieval for optically thin than for thick clouds, the negative bias of the COT ratio in Fig. 9(a) is smaller than that in Fig. 9(b). The negative biases applied to the CTT and SST in the simulation process lead to an underestimation of brightness temperature, which can result in a higher COT retrieval result from the TIR method, thereby inducing negative biases in the simulated COT ratio. However, the COT ratio from an observation is generally >1, when SZA > 20°, so potential biases in the VIS-NIR method might contribute to positive biases in the COT ratio.

As SZA and VZA increase, the impact of the cloud 3-D radiative effect on COT retrievals from the VIS-NIR method becomes significant. For the pixels covered by inhomogeneous clouds with a large SZA, the retrieved COT from the VIS-NIR method tends to have larger values as VZA increases [56]. At two different ranges of VZA, Grosvenor and Wood [57] show a consistent increasing trend of COT with SZA. Several other studies, such as Loeb and Davies [58] and Loeb and Coakley [59], have also reported a strong correlation between COT and SZA using observations from the Earth Radiation Budget Satellite and Advanced Very High Resolution Radiometer. By using a Monte Carlo 3-D radiative transfer model, Loeb et al. [60] demonstrate that the reflectances increase with SZA due to the cloud 3-D radiative effect within a heterogeneous cloud, which is consistent with the previous observation results. Therefore, as shown in Fig. 4(d)–(f), the cloud 3-D radiative effect and cloud inhomogeneity effect are likely to play a dominant role in the overestimation of VIS-NIR COT retrievals. This potential overestimation subsequently would lead to the positive angular dependence of the COT ratio across scattering angles when SZA > 20°. Iwabuchi [61] developed an efficient 3-D radiative transfer model that has been widely used for investigating 3-D effects on cloud optical property retrievals [62], [63], [64]. In future studies, it will be applied in the retrieval system to quantitatively examine the impact of cloud 3-D structure on the angular dependence of the COT ratio.

V. Conclusion

This study examined the spectral consistency of the COT values retrieved from the VIS-NIR and TIR techniques, with a focus on the potential impacts of mixed-phase clouds, CTT, and SST. The ratio of COTs retrieved from the VIS-NIR and TIR retrieval methods displays a significant dependence on scattering angle when clouds are optically thin and have cold cloud top temperatures. The COT ratio distributions exhibit distinct differences in two different SZA ranges.

The COT ratio from observations generally remains below one when SZA < 20°, but exhibits an increasing trend with scattering angle ranging from 100° to 140°. Based on the mixed-phase model and THM, the synthetic simulation results of the COT ratio are similar to the observation results. This indicates that mixed-phase clouds affect the spectral consistency of the COT retrievals, as the variation of the COT ratio reflects the disparity between $\langle P_{11} \rangle$ of the THM and mixed-phase models. As the ice-cloud COT increases, the impact of $\langle g \rangle$ becomes more significant compared to $\langle g \rangle$ due to increased multiple scattering. Therefore, for SZA < 20°, the simulated COT ratio varies less, and its peak ratio near the scattering angle 140° decreases as COT increases. This behavior is consistent with the COT ratio from observations.

When SZA ranges from 20° to 60°, most retrieved COT ratios exceed 1.0 and vary irregularly with scattering angle. The variation cannot be explained by the presence of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:58 UTC from IEEE Xplore. Restrictions apply.

mixed-phase clouds. COT from the TIR retrievals could also be biased due to potential errors in CTT and SST. A simulation of the TIR retrievals revealed that negatively biased CTTs and SSTs lead to negative biases in the COT ratio that are essentially constant with respect to the scattering angle. Thus, the TIR retrieval biases cannot explain observed positive biases in the COT ratio. Previous studies suggest that COT retrievals based on VIS-NIR are overestimated when SZA or VZA is large. These biases may be the result of the cloud 3-D radiative effect and cloud inhomogeneity effect, which could contribute to the positive angular-dependent biases observed in the COT ratio from satellite observations.

Atmospheric and surface properties can introduce biases in existing methods for retrieving the cloud thermodynamic phase based on passive remote sensing instruments, leading to ambiguous results [65]. Active sensors potentially may provide more accurate cloud-top phase retrievals, and have been widely used to validate cloud-top phase retrievals based on observations made by space-based passive sensors [50], [66], [67], [68], [69]. However, the accuracy of identifying mixed-phase clouds by using active sensors is still challenging [70], [71]. By analyzing the angular variation of the COT ratios calculated from VIS-NIR and TIR retrieval methods, it appears that we can efficiently detect the presence of mixed-phase clouds on a large spatial scale with high temporal resolution. This analysis provides a valuable clue for diagnosing the cloud phase and quantitatively evaluating the impact of mixed-phase or liquid clouds on ice cloud retrievals.

ACKNOWLEDGMENT

The authors would like to thank NOAA Comprehensive Large Array-data Stewardship System (CLASS) for providing Geostationary Operational Environmental Satellite (GOES-R) series Advanced Baseline Imager (ABI) products (https://www.avl.class.noaa.gov), Modern-Era Retrospective Analysis for Research and Applications, version 2 (MERRA-2) for providing assimilation products (https://disk.gsfc.nasa.gov), and Jet Propulsion Laboratory (JPL) Physical Oceanography Distributed Active Archive Center (PO-DAAC) for providing GHRSST Level 4 sea surface temperature (SST) products (https://www.ncei.noaa.gov/archive/accession/GHRSST-MUR25-JPL-L4-GLOB). They would also like to thank Dr. Steven Schroder for helping improve the proficiency of writing in the English language in this article. The numerical computations were conducted with high-performance computing resources provided by Texas A&M University, College Station, TX, USA.

REFERENCES

[1] K.-N. Liou and P. Yang, *Light Scattering by Ice Crystals: Fundamentals and Applications*. Cambridge, U.K.: Cambridge Univ. Press, 2016.

[2] D. E. Waliser et al., "Cloud ice: A climate model challenge with signs and expectations of progress," *J. Geophys. Res., Atmos.*, vol. 114, no. D8, Apr. 2009, Art. no. D00A21.

[3] H. Wei et al., "Retrieval of semitransparent ice cloud optical thickness from atmospheric infrared sounder (AIRS) measurements," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 10, pp. 2254–2267, Oct. 2004, doi: 10.1109/TGRS.2004.833780.

[4] F. Chang and Z. Li, "Estimating the vertical variation of cloud droplet effective radius using multispectral near-infrared satellite measurements," *J. Geophys. Res., Atmos.*, vol. 107, no. D15, pp. 1–12, Aug. 2002.

[5] M.-L.-C. Wu, "A method for remote sensing the emissivity, fractional cloud cover and cloud top temperature of high-level, thin clouds," *J. Climate Appl. Meteorol.*, vol. 26, no. 2, pp. 225–233, Feb. 1987.

[6] B. A. Baum and J. D. Spinhirne, "Remote sensing of cloud properties using MODIS airborne simulator imagery during SUCCESS: 3. Cloud overlap," *J. Geophys. Res., Atmos.*, vol. 105, no. D9, pp. 11793–11804, May 2000, doi: 10.1029/1999jd901091.

[7] C. Wang, S. Platnick, Z. Zhang, K. Meyer, G. Wind, and P. Yang, "Retrieval of ice cloud properties using an optimal estimation algorithm and MODIS infrared observations: 2. Retrieval evaluation," *J. Geophys. Res., Atmos.*, vol. 121, no. 10, pp. 5827–5845, May 2016.

[8] H. Iwabuchi et al., "Cloud property retrieval from multiband infrared measurements by Himawari-8," *J. Meteorolog. Soc. Japan. Ser. II*, vol. 96B, pp. 27–42, Jan. 2018.

[9] M. Saito et al., "An efficient method for microphysical property retrievals in vertically inhomogeneous marine water clouds using MODIS-CloudSat measurements," *J. Geophys. Res., Atmos.*, vol. 124, no. 4, pp. 2174–2193, Feb. 2019, doi: 10.1029/2018jd029659.

[10] P. Minnis et al., "CERES MODIS cloud product retrievals for edition 4—Part I: Algorithm changes," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 2744–2780, Apr. 2021.

[11] P. Minnis et al., "CERES edition-2 cloud property retrievals using TRMM VIRS and Terra and Aqua MODIS data—Part I: Algorithms," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4374–4400, 2011.

[12] S. Platnick et al., "The MODIS cloud optical and microphysical products: Collection 6 updates and examples from Terra and Aqua," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 1, pp. 502–525, Jan. 2017.

[13] T. Nakajima and M. D. King, "Determination of the optical thickness and effective particle radius of clouds from reflected solar radiation measurements. Part I: Theory," *J. Atmos. Sci.*, vol. 47, no. 15, pp. 1878–1893, Aug. 1990.

[14] T. Inoue, "On the temperature and effective emissivity determination of semi-transparent cirrus clouds by bi-spectral measurements in the 10&μm window region," *J. Meteorolog. Soc. Jpn. Ser. II*, vol. 63, no. 1, pp. 88–99, 1985.

[15] H. Iwabuchi, S. Yamada, S. Katagiri, P. Yang, and H. Okamoto, "Radiative and microphysical properties of cirrus cloud inferred from infrared measurements made by the Moderate Resolution Imaging Spectroradiometer (MODIS). Part I: Retrieval method," *J. Appl. Meteorol. Climatol.*, vol. 53, no. 5, pp. 1297–1316, May 2014, doi: 10.1175/jamc-d-13-0215.1.

[16] A. Macke, J. Mueller, and E. Raschke, "Single scattering properties of atmospheric ice crystals," *J. Atmos. Sci.*, vol. 53, no. 19, pp. 2813–2825, Oct. 1996.

[17] H. Eichler et al., "Influence of ice crystal shape on retrieval of cirrus optical thickness and effective radius: A case study," *J. Geophys. Res., Atmos.*, vol. 114, no. D19, pp. 1–10, Oct. 2009, doi: 10.1029/2009jd012215.

[18] B. A. Baum et al., "Improvements in shortwave bulk scattering and absorption models for the remote sensing of ice clouds," *J. Appl. Meteorol. Climatol.*, vol. 50, no. 5, pp. 1037–1056, May 2011, doi: 10.1175/2010jamc2608.1.

[19] N. G. Loeb et al., "Impact of ice cloud microphysics on satellite cloud retrievals and broadband flux radiative transfer model calculations," *J. Climate*, vol. 31, no. 5, pp. 1851–1864, Mar. 2018, doi: 10.1175/jcli-d-17-0426.1.

[20] P. N. Francis, J. S. Foot, and A. J. Baran, "Aircraft measurements of the solar and infrared radiative properties of cirrus and their dependence on ice crystal shape," *J. Geophys. Res. Atmos.*, vol. 104, no. D24, pp. 31685–31695, 1999, doi: 10.1029/1999JD900438.

[21] M. Saito and P. Yang, "Oriented ice crystals: A single-scattering property database for applications to LiDAR and optical phenomenon simulations," *J. Atmos. Sci.*, vol. 76, no. 9, pp. 2635–2652, Sep. 2019.

[22] M. Li et al., "Retrieval of terahertz ice cloud properties from airborne measurements based on the irregularly shaped Voronoi ice scattering models," *Atmos. Meas. Techn.*, vol. 16, no. 2, pp. 331–353, Jan. 2023.

[23] J. J. Coy, M. Saito, P. Yang, X. Liu, and Y. Hu, "An improved ice cloud optical property model for lidar-based remote sensing applications," *IEEE Geosci. Remote Sens. Lett.*, 2023.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:58 UTC from IEEE Xplore. Restrictions apply.

[24] P. Yang et al., "Spectrally consistent scattering, absorption, and polarization properties of atmospheric ice crystals at wavelengths from 0.2 to 100 μm," *J. Atmos. Sci.*, vol. 70, no. 1, pp. 330–347, Jan. 2013.

[25] A. Heidinger, Y. Li, B. Baum, R. Holz, S. Platnick, and P. Yang, "Retrieval of cirrus cloud optical depth under day and night conditions from MODIS collection 6 cloud property data," *Remote Sens.*, vol. 7, no. 6, pp. 7257–7271, Jun. 2015, doi: 10.3390/rs70607257.

[26] P. Yang, S. Hioki, M. Saito, C.-P. Kuo, B. Baum, and K.-N. Liou, "A review of ice cloud optical property models for passive satellite remote sensing," *Atmosphere*, vol. 9, no. 12, p. 499, Dec. 2018, doi: 10.3390/atmos9120499.

[27] Z. Zhang et al., "A framework based on 2-D Taylor expansion for quantifying the impacts of subpixel reflectance variance and covariance on cloud optical thickness and effective radius retrievals based on the bispectral method," *J. Geophys. Res., Atmos.*, vol. 121, no. 12, pp. 7007–7025, Jun. 2016.

[28] D. Li, M. Saito, and P. Yang, "Time-dependent systematic biases in inferring ice cloud properties from geostationary satellite observations," *Remote Sens.*, vol. 15, no. 3, p. 855, Feb. 2023, doi: 10.3390/rs15030855.

[29] F. Werner, Z. Zhang, G. Wind, D. J. Miller, and S. Platnick, "Quantifying the impacts of subpixel reflectance variability on cloud optical thickness and effective radius retrievals based on high-resolution ASTER observations," *J. Geophys. Res., Atmos.*, vol. 123, no. 8, pp. 4239–4258, Apr. 2018, doi: 10.1002/2017jd027916.

[30] T. Zinner and B. Mayer, "Remote sensing of stratocumulus clouds: Uncertainties and biases due to inhomogeneity," *J. Geophys. Res., Atmos.*, vol. 111, no. D14, Jul. 2006, Art. no. D14209.

[31] T. Fauchez et al., "Impacts of cloud heterogeneities on cirrus optical properties retrieved from space-based thermal infrared radiometry," *Atmos. Meas. Techn.*, vol. 8, no. 2, pp. 633–647, Feb. 2015, doi: 10.5194/amt-8-633-2015.

[32] B. Marchant, S. Platnick, K. Meyer, and G. Wind, "Evaluation of the MODIS collection 6 multilayer cloud detection algorithm through comparisons with CloudSat cloud profiling radar and CALIPSO CALIOP products," *Atmos. Meas. Techn.*, vol. 13, no. 6, pp. 3263–3275, Jun. 2020, doi: 10.5194/amt-13-3263-2020.

[33] C. A. Poulsen et al., "Cloud retrievals from satellite data using optimal estimation: Evaluation and application to ATSR," *Atmos. Meas. Techn.*, vol. 5, no. 8, pp. 1889–1910, Aug. 2012, doi: 10.5194/amt-5-1889-2012.

[34] Y. Zhou et al., "Cirrus cloud optical thickness and effective diameter retrieved by MODIS: Impacts of single habit assumption, 3-D radiative effects, and cloud inhomogeneity," *J. Geophys. Res., Atmos.*, vol. 123, no. 2, pp. 1195–1210, Jan. 2018, doi: 10.1002/2017jd027232.

[35] P. Minnis, J. M. Alvarez, K. Sassen, D. F. Young, and C. J. Grund, "The 27–28 October 1986 FIRE IFO cirrus case study: Cirrus parameter relationships derived from satellite and LiDAR data," *Monthly Weather Rev.*, vol. 118, no. 11, pp. 2402–2425, Nov. 1990.

[36] H.-M. Cho, S. L. Nasiri, and P. Yang, "Application of CALIOP measurements to the evaluation of cloud phase derived from MODIS infrared channels," *J. Appl. Meteorol. Climatol.*, vol. 48, no. 10, pp. 2169–2180, Oct. 2009.

[37] S. L. Nasiri and B. H. Kahn, "Limitations of bispectral infrared cloud phase determination and potential for improvement," *J. Appl. Meteorol. Climatol.*, vol. 47, no. 11, pp. 2895–2910, Nov. 2008.

[38] B. A. Baum et al., "MODIS cloud-top property refinements for collection 6," *J. Appl. Meteorol. Climatol.*, vol. 51, no. 6, pp. 1145–1163, Jun. 2012, doi: 10.1175/jamc-d-11-0203.1.

[39] S. Hioki et al., "Degree of ice particle surface roughness inferred from polarimetric observations," *Atmos. Chem. Phys.*, vol. 16, no. 12, pp. 7545–7558, Jun. 2016, doi: 10.5194/acp-16-7545-2016.

[40] B. P. Briegleb, P. Minnis, V. Ramanathan, and E. Harrison, "Comparison of regional clear-sky albedos inferred from satellite observations and model computations," *J. Climate Appl. Meteorol.*, vol. 25, no. 2, pp. 214–226, Feb. 1986.

[41] G. Mie, "Beiträge zur optik trüber medien, speziell kolloidaler Metallösungen," *Annalen der Physik*, vol. 330, no. 3, pp. 377–445, Jan. 1908.

[42] M. Saito and P. Yang, "Advanced bulk optical models linking the backscattering and microphysical properties of mineral dust aerosol," *Geophys. Res. Lett.*, vol. 48, no. 17, Sep. 2021, Art. no. e2021GL095121.

[43] P. Yang et al., "The spectral signature of mixed-phase clouds composed of non-spherical ice crystals and spherical liquid droplets in the terrestrial window region," *J. Quant. Spectrosc. Radiat. Transf.*, vols. 79–80, pp. 1171–1188, Jun. 2003.

[44] X. Huang, P. Yang, G. Kattawar, and K.-N. Liou, "Effect of mineral dust aerosol aspect ratio on polarized reflectance," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 151, pp. 97–109, Jan. 2015.

[45] M. Saito, P. Yang, A. K. Heidinger, and Y. Li, "An improved beta method for ice cloud property retrievals: Theory," *J. Geophys. Res., Atmos.*, vol. 125, no. 14, Jul. 2020, Art. no. e2019JD031863.

[46] M. Saito, P. Yang, X. Huang, H. E. Brindley, M. G. Mlynczak, and B. H. Kahn, "Spaceborne middle- and far-infrared observations improving nighttime ice cloud property retrievals," *Geophys. Res. Lett.*, vol. 47, no. 18, Sep. 2020, Art. no. e2020GL087491.

[47] C. D. Rodgers, *Inverse Methods for Atmospheric Sounding: Theory and Practice*, vol. 2. Singapore: World Scientific, 2000.

[48] M. Pavolonis and C. Calvert, "GOES-R R advanced baseline imager (ABI) algorithm theoretical basis document for cloud type and cloud phase," NOAA/NESDIS/STAR, MD, USA, 2020.

[49] J. Hu, D. Rosenfeld, Y. Zhu, X. Lu, and J. Carlin, "Multi-channel imager algorithm (MIA): A novel cloud-top phase classification algorithm," *Atmos. Res.*, vol. 261, Oct. 2021, Art. no. 105767, doi: 10.1016/j.atmosres.2021.105767.

[50] E. L. A. Wolters, R. A. Roebeling, and A. J. Feijt, "Evaluation of cloud-phase retrieval methods for SEVIRI on Meteosat-8 using ground-based LiDAR and cloud radar data," *J. Appl. Meteorol. Climatol.*, vol. 47, no. 6, pp. 1723–1738, Jun. 2008.

[51] Y. Huang, S. Siems, M. Manton, A. Protat, L. Majewski, and H. Nguyen, "Evaluating Himawari-8 cloud products using shipborne and CALIPSO observations: Cloud-top height and cloud-top temperature," *J. Atmos. Ocean. Technol.*, vol. 36, no. 12, pp. 2327–2347, Dec. 2019.

[52] U. Hamann et al., "Remote sensing of cloud top pressure/height from SEVIRI: Analysis of ten current retrieval algorithms," *Atmos. Meas. Techn.*, vol. 7, no. 9, pp. 2839–2867, Sep. 2014.

[53] S. L. Ermida, I. F. Trigo, C. C. DaCamara, C. Jiménez, and C. Prigent, "Quantifying the clear-sky bias of satellite Land Surface Temperature using microwave-based estimates," *J. Geophys. Res., Atmos.*, vol. 124, no. 2, pp. 844–857, Jan. 2019, doi: 10.1029/2018jd029354.

[54] C. B. Lima, S. S. Prijith, M. V. R. S. Sai, P. V. N. Rao, K. Niranjan, and M. V. Ramana, "Retrieval and validation of cloud top temperature from the geostationary satellite INSAT-3D," *Remote Sens.*, vol. 11, no. 23, p. 2811, Nov. 2019.

[55] A. Heidinger, Y. Li, and S. Wanzong, "Enterprise AWG cloud height algorithm (ACHA)," NOAA/NESDIS/STAR, MD, USA, 2020.

[56] T. Várnai and A. Marshak, "View angle dependence of cloud optical thicknesses retrieved by Moderate Resolution Imaging Spectroradiometer (MODIS)," *J. Geophys. Res., Atmos.*, vol. 112, no. D6, pp. 1–12, Mar. 2007, doi: 10.1029/2005jd006912.

[57] D. P. Grosvenor and R. Wood, "The effect of solar zenith angle on MODIS cloud optical and microphysical retrievals within marine liquid water clouds," *Atmos. Chem. Phys.*, vol. 14, no. 14, pp. 7291–7321, Jul. 2014, doi: 10.5194/acp-14-7291-2014.

[58] N. G. Loeb and R. Davies, "Observational evidence of plane parallel model biases: Apparent dependence of cloud optical depth on solar zenith angle," *J. Geophys. Res., Atmos.*, vol. 101, no. D1, pp. 1621–1634, Jan. 1996.

[59] N. G. Loeb and J. A. Coakley, "Inference of marine stratus cloud optical depths from satellite measurements: Does 1D theory apply?" *J. Clim.*, vol. 11, no. 2, pp. 215–233, 1998, doi: 10.1175/1520-0442(1998)011<0215:IOMSCO>2.0.CO.2.

[60] N. G. Loeb, T. Várnai, and R. Davies, "Effect of cloud inhomogeneities on the solar zenith angle dependence of nadir reflectance," *J. Geophys. Res., Atmos.*, vol. 102, no. D8, pp. 9387–9395, Apr. 1997.

[61] H. Iwabuchi, "Efficient Monte Carlo methods for radiative transfer modeling," *J. Atmos. Sci.*, vol. 63, no. 9, pp. 2324–2339, Sep. 2006.

[62] R. Masuda, H. Iwabuchi, K. S. Schmidt, A. Damiani, and R. Kudo, "Retrieval of cloud optical thickness from sky-view camera images using a deep convolutional neural network based on three-dimensional radiative transfer," *Remote Sens.*, vol. 11, no. 17, p. 1962, Aug. 2019.

[63] H. Iwabuchi and T. Hayasaka, "Effects of cloud horizontal inhomogeneity on the optical thickness retrieved from moderate-resolution satellite data," *J. Atmos. Sci.*, vol. 59, no. 14, pp. 2227–2242, Jul. 2002.

[64] H. Iwabuchi and T. Hayasaka, "A multi-spectral non-local method for retrieval of boundary layer cloud properties from optical remote sensing data," *Remote Sens. Environ.*, vol. 88, no. 3, pp. 294–308, Dec. 2003.

[65] J. Riedi et al., "Cloud thermodynamic phase inferred from merged POLDER and MODIS data," *Atmos. Chem. Phys.*, vol. 10, no. 23, pp. 11851–11865, Dec. 2010.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:58 UTC from IEEE Xplore. Restrictions apply.

[66] O. Bruno, C. Hoose, T. Storelvmo, Q. Coopman, and M. Stengel, "Exploring the cloud top phase partitioning in different cloud types using active and passive satellite sensors," *Geophys. Res. Lett.*, vol. 48, no. 2, Jan. 2021, Art. no. e2020GL089863.
[67] E. Jäkel, J. Walter, and M. Wendisch, "Thermodynamic phase retrieval of convective clouds: Impact of sensor viewing geometry and vertical distribution of cloud properties," *Atmos. Meas. Techn.*, vol. 6, no. 3, pp. 539–547, Mar. 2013.
[68] T. M. Nagao and K. Suzuki, "Temperature-independent cloud phase retrieval from shortwave-infrared measurement of GCOM/C/SGLI with comparison to CALIPSO," *Earth Space Sci.*, vol. 8, no. 11, Nov. 2021, Art. no. e2021EA001912.
[69] S. Zeng, J. Riedi, F. Parol, C. Cornet, and F. Thieuleux, "An assessment of cloud top thermodynamic phase products obtained from A-train passive and active sensors," *Atmos. Meas. Tech. Discuss.*, vol. 6, no. 5, pp. 8371–8411, 2013.
[70] A. V. Matus and T. S. L'Ecuyer, "The role of cloud phase in Earth's radiation budget," *J. Geophys. Res., Atmos.*, vol. 122, no. 5, pp. 2559–2578, Mar. 2017.
[71] G. Cesana et al., "Using in situ airborne measurements to evaluate three cloud phase products derived from CALIPSO," *J. Geophys. Res., Atmos.*, vol. 121, no. 10, pp. 5788–5808, May 2016.



Dongchen Li is currently a Graduate Research Assistant with Texas A&M University, College Station, TX, USA. His research interests include satellite remote sensing of clouds and aerosols, and radiative transfer.



Masanori Saito received the B.S., M.S., and Ph.D. degrees in geophysics from Tohoku University, Sendai, Japan, in 2012, 2014, and 2017, respectively.

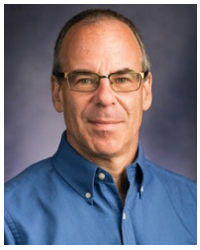
He recently joined the Department of Atmospheric Science, University of Wyoming, Laramie, WY, USA, as an Assistant Professor, after he spent six years as a Post-Doctoral Research Associate and Assistant Research Scientist with Texas A&M University, College Station, TX, USA. His research interests include the characterization of cloud and aerosol properties using remote sensing techniques.



Ping Yang (Fellow, IEEE) received the B.S. degree in theoretical physics and from Lanzhou University, Lanzhou, China, in 1985, the M.S. degree in atmospheric physics from the Lanzhou Institute of Plateau Atmospheric Physics, Chinese Academy of Sciences, Lanzhou, in 1988, and the Ph.D. degree in meteorology from the University of Utah, Salt Lake City, UT, USA, in 1995.

He is currently a University Distinguished Professor and a Senior Associate Dean for Research and Graduate Studies with the College of Arts and Sciences, Texas A&M University, College Station, TX, USA, and the David Bullock Harris Chair in Geosciences. His research interests include light scattering, radiative transfer, and remote sensing. He has coauthored 368 referred journal articles and four books.

Dr. Yang is a Fellow of the American Meteorological Society (AMS), the American Geophysical Union (AGU), the American Physical Society (APS), OPTICA (formerly, the Optical Society of America, OSA), the Electromagnetics Academy, and the American Association for the Advancement of Science (AAAS).



Norman G. Loeb received the B.Sc. degree from McGill University, Montreal, QC, Canada, in 1985, the M.Sc. degree from York University, Toronto, ON, Canada, in 1987, and the Ph.D. degree in atmospheric physics from McGill University in 1996.

He is currently a Senior Scientist for radiation sciences with the NASA Langley Research Center, Hampton, VA, USA, and a Principal Investigator of the Clouds and the Earth's Radiant Energy System (CERES) Project. His research interests include Earth radiation budget, remote sensing, and climate variability and change.



William L. Smith Jr. received the B.S. degree in meteorology from the University of Wisconsin-Madison, Madison, WI, USA, in 1986, the M.S. degree in atmospheric science from Colorado State University, Fort Collins, CO, USA, in 1990, and the Ph.D. degree in atmospheric and oceanic sciences from the University of Wisconsin-Madison in 2014.

He is currently a Scientist with the Climate Science Branch, Science Directorate, NASA Langley Research Center, Hampton, VA, USA, where he uses satellite data to study clouds and radiation and develops data products for weather and climate applications.



Patrick Minnis received the B.E. degree in materials science and metallurgical engineering from Vanderbilt University, Nashville, TN, USA, in 1972, the M.S. degree in atmospheric science from Colorado State University, Fort Collins, CO, USA, in 1978, and the Ph.D. degree in meteorology from the University of Utah, Salt Lake City, UT, USA, in 1991.

He retired from the Climate Science Branch, NASA Langley Research Center, Hampton, VA, USA, where he was a Senior Research Scientist from 1981 to 2017. He is currently a Consulting Research Scientist with Analytical Mechanics Associates Inc., Hampton. He has authored or coauthored over 310 peer reviewed publications. His research is focused on the remote sensing of clouds and surface properties from satellite imagery for weather and climate investigations. He supports the NASA CERES Project as part of a research group that also analyzes polar-orbiting and geostationary satellite data for field missions, operational aircraft icing condition diagnosis, and contrail research.

Dr. Minnis is a Fellow of the American Geophysical Union and American Meteorological Society. He was a recipient of NASA medals for Exceptional Scientific Achievement in 1993, Exceptional Achievement in 2005, Exceptional Service in 2008, and Distinguished Service in 2011. He received the 2008 AMS Henry G. Houghton Award for Atmospheric Physics, the 2011 AIAA Losey Atmospheric Sciences Award, and the 2019 AMS Verner E. Suomi Technology Medal.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:58 UTC from IEEE Xplore. Restrictions apply.